Combined Families Summary

LLM · ASR · TTS bake-off — cross-family overview

Prepared 23 July 2026 · Egyptian Arabic voice robot stack

LLM (speed)	LLM (quality)	ASR pick	TTS pick
Nile-Chat-4B	**Qwen3-8B**	**Whisper-Large-v3-Turbo-CT2**	**VoiceTut-TTS**

Executive verdict

Recommended package: Whisper-Large-v3-Turbo-CT2 + Nile-Chat-4B + VoiceTut-TTS (~13.7 GB peak VRAM sum). Quality package: Whisper-Large-v3-Turbo-CT2 + Qwen3-8B + VoiceTut-TTS (~12.2

Overview

This summary helps select one model from each family for the Arabic voice robot stack.
All three families now have successful bake-off results and shortlists.
Suggested ASR default: Whisper-Large-v3-Turbo-CT2. Accuracy alternative: Whisper-Large-v3-CT2.
Suggested LLM speed default: Nile-Chat-4B. Quality alternative: Qwen3-8B.
Suggested TTS default (speed/resources): VoiceTut-TTS - confirm by listening to the WAV.
Prefer LOWER latency/memory/error rates; prefer HIGHER accuracy/throughput/quality/robot scores.

How to read this report

Page 2: shared metrics glossary.
Page 3: shortlist table and decision path.
Page 4: combined GPU / VRAM stack budget (critical).
Page 5: family readiness and top scores.
Pages 6-8: LLM, ASR, and TTS snapshot charts.
Page 9: package contents and bottom line.

Combined - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

TTFT - Time To First Token (s)

LOWER better
Delay until the model starts generating. Lower = snappier turn-taking (robot starts sooner).

Throughput (tokens/s)

HIGHER better
How fast tokens are generated after the first one. Higher = long answers finish sooner.

Generate time (s)

LOWER better
Total time to finish the reply. Lower is better (also depends on answer length).

Accuracy % (word accuracy)

HIGHER better
Share of words recognized correctly (about 1 - WER). Higher = fewer recognition mistakes.

WER - Word Error Rate

LOWER better
Fraction of words wrong. 0 = perfect. Lower = better recognition.

CER - Character Error Rate

LOWER better
Same idea as WER at character level. Useful for Arabic. Lower = better.

Chars/s (characters per second)

HIGHER better
Input characters synthesized per second of compute. Higher = faster speech generation.

Audio length (s)

Context only

Peak VRAM (MB)

LOWER better
GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.

Peak RAM (MB)

LOWER better
System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better
CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better
Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better
Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.

x Realtime (xRT)

HIGHER better
How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.

Robot realtime score (0-100)

HIGHER better
Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Success rate %

HIGHER better

Remember

Shared rule: LOWER = TTFT, RTF, WER, CER, VRAM, RAM, CPU%, load/generate/transcribe time.
Shared rule: HIGHER = accuracy%, tokens/s, chars/s, quality scores, robot/balanced scores, xRealtime.
See each family PDF glossary for the full term list.

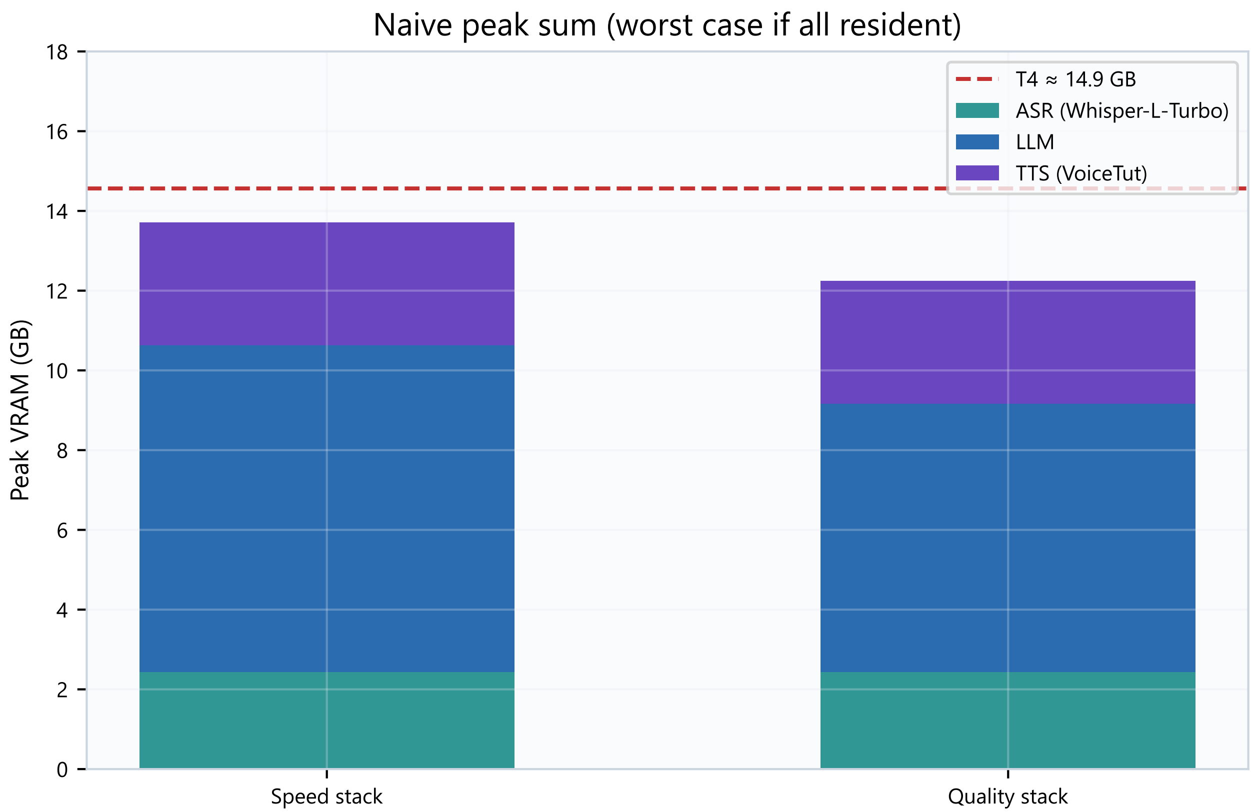
Recommended shortlist

ASR — robot default **Whisper-Large-v3-Turbo-CT2**	Best accuracy/speed/VRAM blend for realtime input
ASR — accuracy alt **Whisper-Large-v3-CT2**	Highest word accuracy when recognition quality is critical
LLM — speed default **Nile-Chat-4B**	Best TTFT/throughput/VRAM composite for turn-taking
LLM — quality alt **Qwen3-8B**	Best rubric quality (4.65/5) and auto-pass rate
TTS — speed/resources **VoiceTut-TTS**	Best RTF/throughput/VRAM composite — listen to WAV before freeze

Decision path

1) Lock ASR first — input quality gates the whole pipeline.
2) Lock LLM next — choose speed (Nile-Chat-4B) or quality (Qwen3-8B / Qwen3-4B).
3) Lock TTS after listening — default VoiceTut-TTS; fallback SILMA-TTS if voice quality wins.
4) Validate the combined VRAM budget on the deployment GPU (next page).

Combined GPU stack budget (peak VRAM)



Budget numbers

ASR: 2.43 GB
LLM speed: 8.20 GB
LLM quality: 6.74 GB
TTS: 3.08 GB

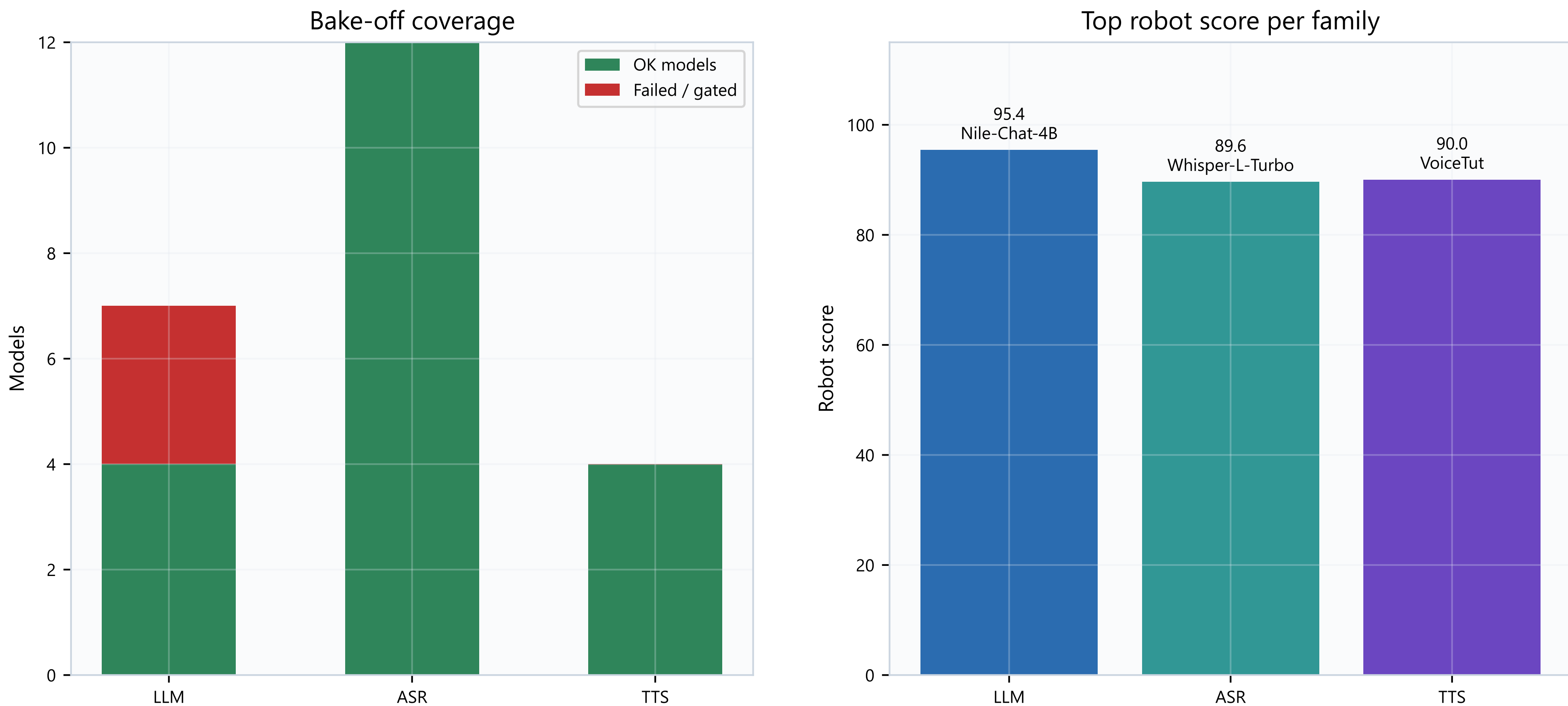
Speed sum: 13.71 GB
Quality sum: 12.25 GB
T4 usable: 14.6 GB

Headroom is tight on speed stack.
Prefer quality LLM (int4) if co-resident.

How to read the stack budget

Speed package = Whisper-Large-v3-Turbo-CT2 + Nile-Chat-4B + VoiceTut-TTS ≈ 13.7 GB peak sum.
Quality package = Whisper-Large-v3-Turbo-CT2 + Qwen3-8B + VoiceTut-TTS ≈ 12.2 GB peak sum.
These are peak-per-family sums — true concurrent residency may be lower if models swap, but plan for the worst case.
On T4-class GPUs, prefer the quality LLM (Qwen3-8B int4) or Whisper-Small if headroom is insufficient.
Always measure actual concurrent VRAM on the deployment box before freeze.

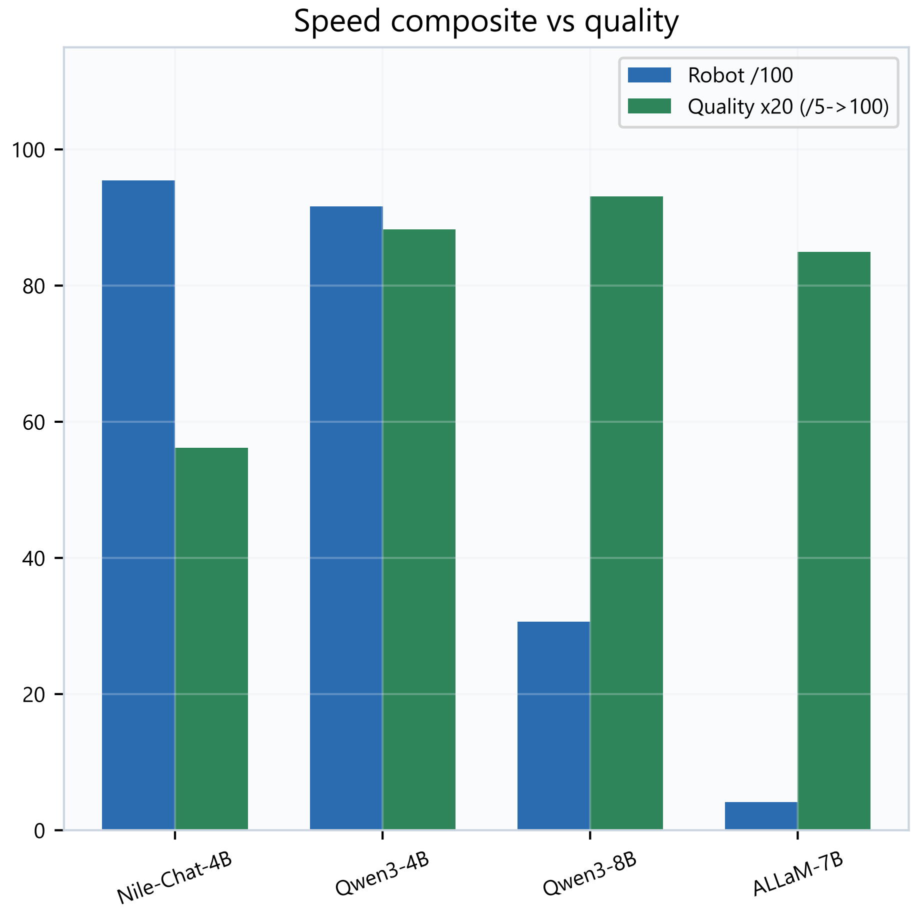
Family readiness & top robot scores

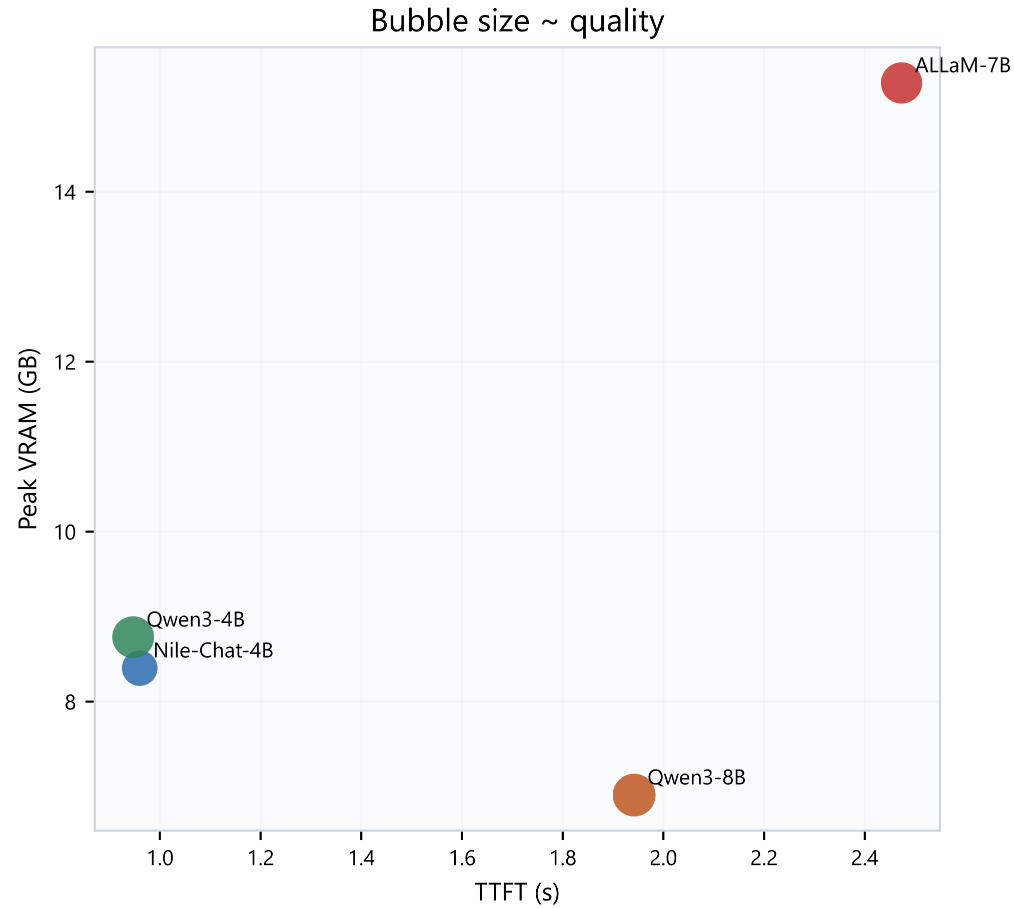


Readiness reading

ASR is fully covered (12/12 OK). LLM has 4 OK models + 3 gated failures.
TTS is ready for shortlisting (4/4 OK) — VoiceTut leads robot score.
All three families can be shortlisted together; only TTS still needs a human listening pass.
Plan GPU headroom for Whisper + LLM + VoiceTut/SILMA together (see stack budget page).

LLM snapshot

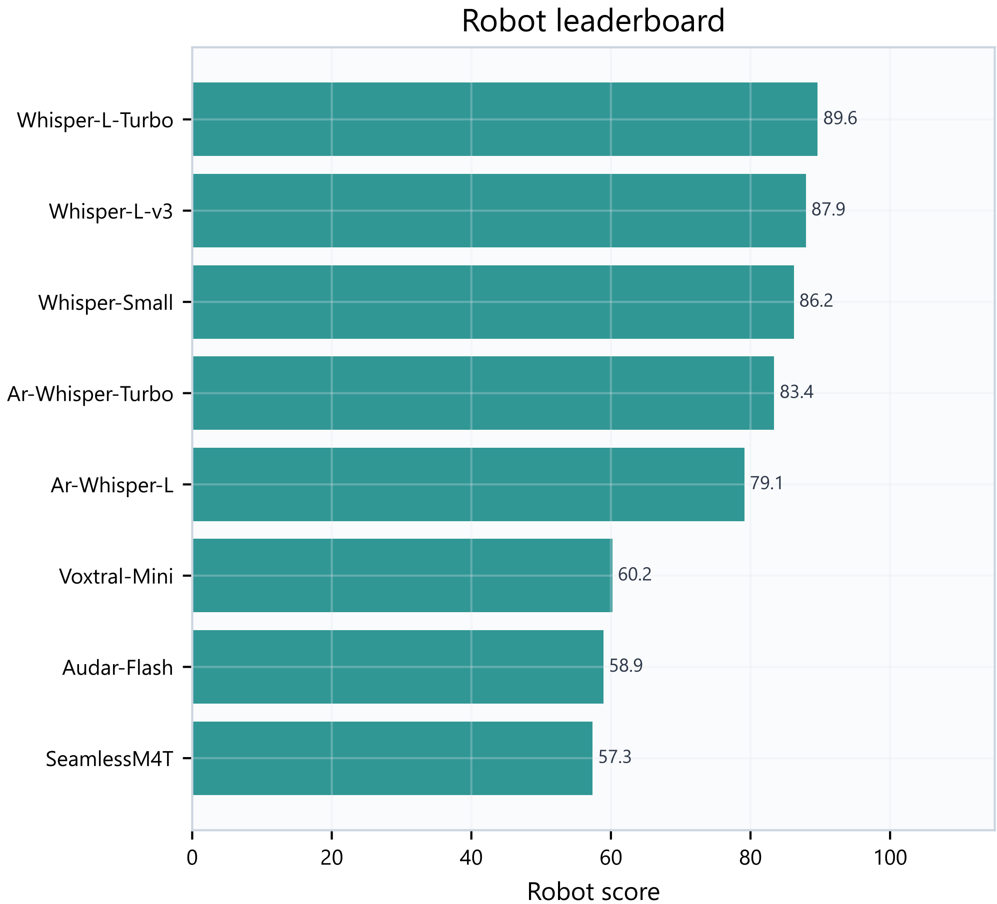


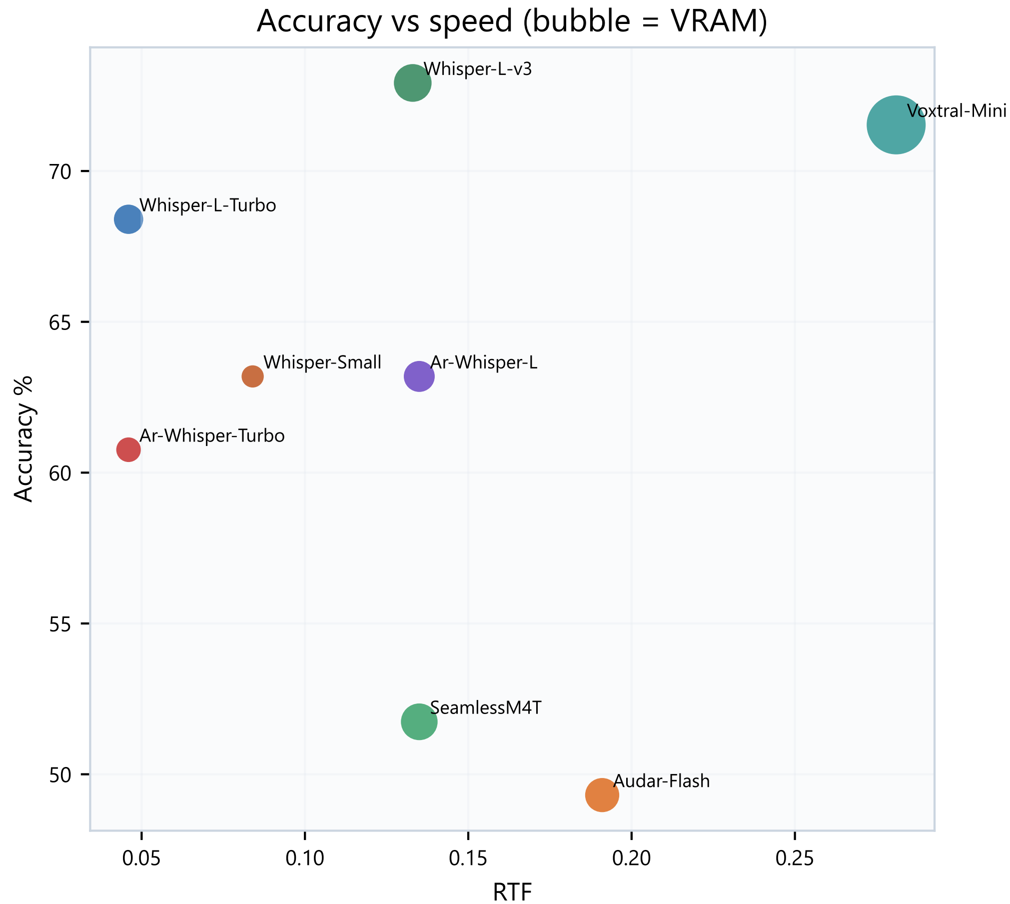


LLM takeaway

Nile-Chat-4B wins robot realtime; Qwen3-8B wins quality.
If spoken answers must stay short and correct, lean quality (Qwen).
If the robot must start speaking immediately, lean Nile-Chat-4B — then spot-check verbosity.
Qwen3-4B is the practical middle ground for many deployments.

ASR snapshot (top 8)

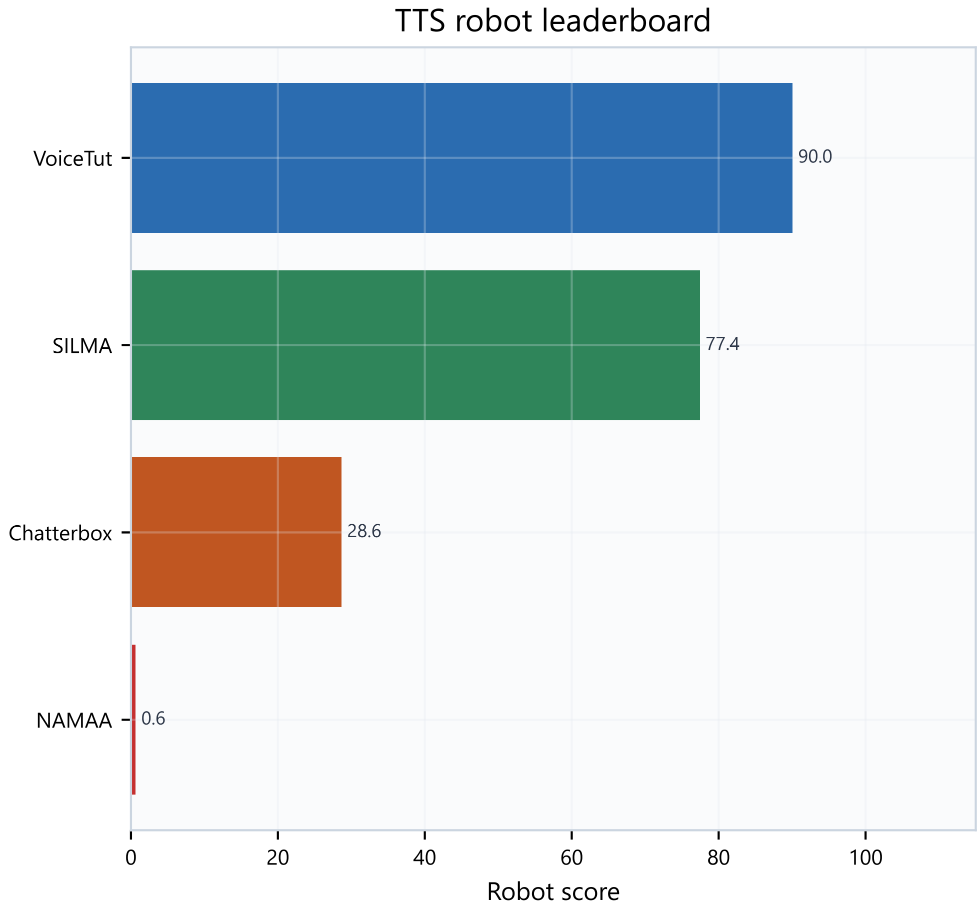


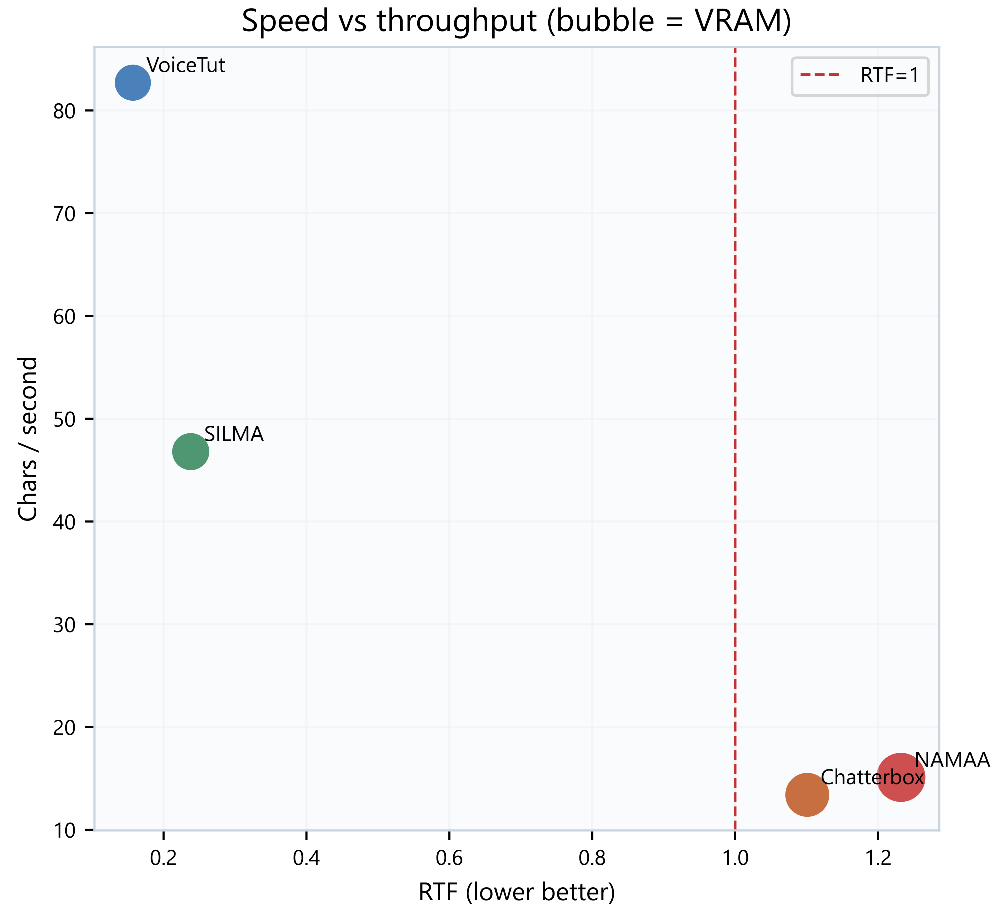


ASR takeaway

Default: Whisper-Large-v3-Turbo-CT2 for robot balance.
Accuracy alternative: Whisper-Large-v3-CT2.
Whisper CT2 family occupies the useful top band; Qwen ASR variants lag badly on WER.
Keep ASR VRAM low so the LLM + TTS still fit on the same GPU.

TTS snapshot





TTS takeaway

Default speed/resources pick: VoiceTut-TTS.
SILMA-TTS is the best realtime alternative if VoiceTut fails the listen test.
Chatterbox and NAMAA are not realtime on this bake-off — use only if voice quality clearly wins.
Always listen to WAVs before freezing the production voice.

Package contents & bottom line

PDF package

LLM_Model_Selection_Report.pdf — methodology + charts + quality guidance
ASR_Model_Selection_Report.pdf — methodology + accuracy/speed evidence
TTS_Model_Selection_Report.pdf — methodology + picks + listening checklist
Combined_Families_Summary.pdf — this cross-family overview + VRAM stack budget

WAV files for listening: analytics/final/tts_outputs/*.wav

Bottom line

ASR = Whisper-Large-v3-Turbo-CT2 (or Whisper-Large-v3-CT2 for max accuracy).
LLM = Nile-Chat-4B (speed) or Qwen3-8B (quality).
TTS = VoiceTut-TTS pending listening confirmation (SILMA as backup).
VRAM plan: speed stack ~13.7 GB · quality stack ~12.2 GB (T4 ~14.9 GB).
See family PDFs for full evidence before final sign-off.